Predicting 5-Year Charge-Off Risk for SBA 7(a) Loans

DATA 400 Individual Mini-Project - Final Report

Fall 2026

Research Question

Using information available around loan approval, how well can we predict whether an SBA 7(a) loan will be charged off within five years?

1. Motivation and Project Goal

Credit risk is a core problem in commercial banking because lenders need to distinguish loans that are more likely to perform from loans that may later produce losses. For this project, I focus on SBA 7(a) small-business loans and ask whether information that is available around the time of approval contains useful predictive information about five-year charge-off risk. I treat the model as a risk-screening tool rather than an automatic approval or rejection rule.

The intended audience includes credit analysts, lenders, portfolio-risk teams, and other users who need to understand patterns in small-business credit performance. The project is predictive, not causal: the goal is to measure how well approval-time loan and business characteristics can help rank future charge-off risk, not to claim that any one variable causes a borrower to fail.

2. Data and Outcome Definition

The data come from the U.S. Small Business Administration 7(a) & 504 FOIA dataset [1]. I use the 7(a) FY2010-FY2019 file, the 7(a) FY2020-Present file, and the official data dictionary. The final source snapshot is June 30, 2026. After exact duplicate removal, the combined source contains 932,013 loan records.

The main outcome is chargeoff_5yr. A loan receives a value of 1 when a documented charge-off occurs within five years after approval and 0 when no documented charge-off occurs during that same five-year window. Recent loans cannot be labeled fairly until five full years have passed, so the modeling sample is restricted to loans with a complete observation window. With a June 30, 2026 snapshot, June 30, 2021 is the latest eligible approval date. After applying the five-year rule, excluding non-funded statuses, and removing 15 cases where charge-off evidence existed but exact timing was unavailable, 543,110 loans remained in the modeling universe. The overall five-year charge-off rate was 3.76%.

Outcome fields such as LoanStatus, ChargeOffDate, GrossChargeOffAmount, PaidInFullDate, and the file as-of date are used only to construct or verify the target. They are not included as predictors because they contain information that would not be available when the original credit decision was made.

Table 1. Chronological modeling split

Split	Approval dates	Rows	5-year charge-off rate
Training	2009-10-01 to 2017-12-31	393,008	3.86%
Validation	2018-01-01 to 2019-12-31	95,853	4.18%
Test	2020-01-01 to 2021-06-30	54,249	2.36%

The training sample starts in October 2009 because the SBA FY2010 source begins at the start of federal fiscal year 2010. I use a chronological split rather than a random split so that model development follows the real use case more closely: learn from older loans, select the model on later loans, and evaluate once on a still-later test period.

3. Data Quality, Features, and Exploratory Analysis

The first stage checks data types, missingness, duplicate rows, unusual values, and loan-status codes. Numeric variables are converted from formatted strings when necessary, dates are standardized, and broad two-digit NAICS sectors are created from the original NAICS codes. I also create SBA guarantee percentage, log loan amount, approval year, and a simple franchise indicator.

The data-quality check found 208 training observations with non-positive loan terms, about 0.05% of the 393,008 training loans. I flag this as a limitation rather than deleting observations after model development. The same checks found no non-positive gross approval amounts, no negative interest rates, and no SBA guarantee percentages above 1.05. IQR rules are used as diagnostics, but statistical outliers are not automatically removed because unusually large loans or long terms can still be legitimate SBA records.

The exploratory analysis showed several clear descriptive patterns. Smaller loans had higher observed five-year charge-off rates: loans below $50,000 had a rate of 5.59%, compared with 0.85% for loans above $1 million. Shorter-term loans also had higher observed rates: loans with terms of 84 months or less had a 5.11% rate, compared with 0.48% for terms above 180 months. Higher-rate loans were riskier in the descriptive data as well; the 8-11% rate band had a 7.30% charge-off rate, compared with 1.35% for rates below 5%. These comparisons describe the sample but do not establish causal effects.

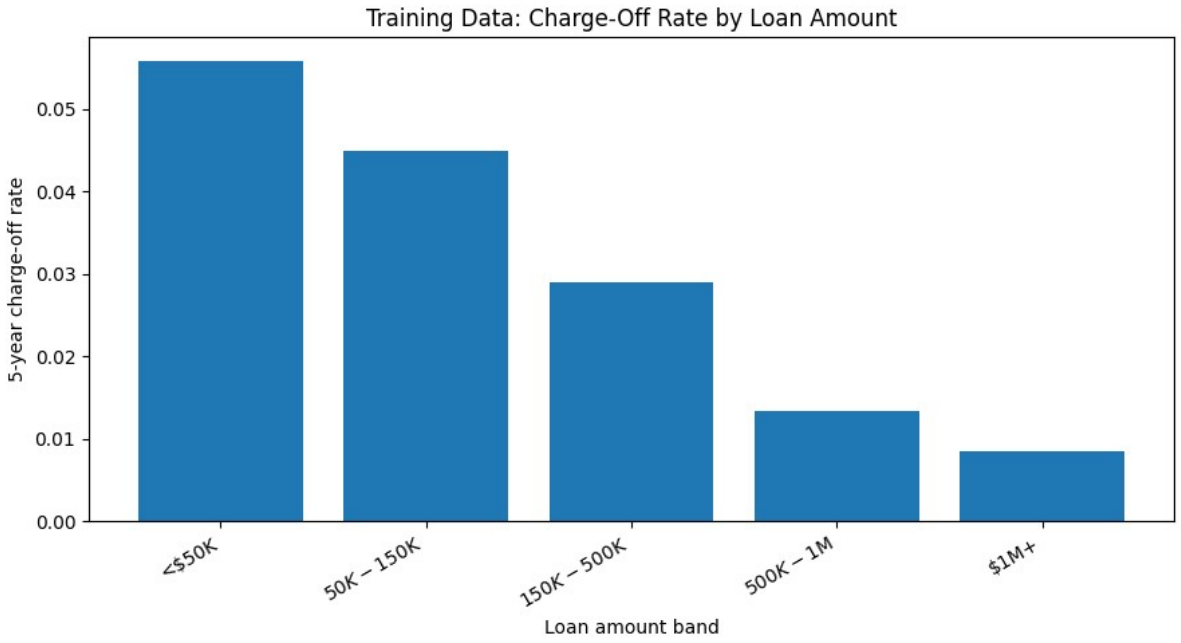


Figure 1. Training-set five-year charge-off rate by loan amount band.

4. Modeling Approach

The modeling inputs are variables that are reasonably available around approval. Numeric predictors include log gross approval amount, SBA guarantee percentage, loan term, initial interest rate, jobs supported, and approval year. Categorical predictors include project state, business type, business age, processing method, fixed/variable rate status, revolving status, collateral indicator, broad NAICS sector, and franchise indicator.

Preprocessing and modeling are combined in Scikit-learn pipelines [4]. Numeric missing values are imputed with the training-set median and standardized. Categorical missing values are imputed with the most frequent category and then one-hot encoded. Keeping preprocessing inside the pipeline helps ensure that transformations are learned from the training data rather than from validation or test information.

I compare four classifiers: a Dummy baseline, Logistic Regression, a Decision Tree, and a Random Forest. Logistic Regression uses balanced class weights and default L2 regularization. The Decision Tree and Random Forest use restricted depth and minimum leaf sizes to control complexity. Random Forest tuning is included as an optional extension in the notebook but is not run in the final analysis because of the large sample and the scope of the mini-project. The final comparison therefore uses the predefined model settings.

Because charge-offs are relatively rare, accuracy is not the main model-selection metric. I use PR-AUC (Average Precision) as the primary comparison measure and also report ROC-AUC. After selecting the best validation model, I choose a single classification threshold on validation data by maximizing F1, which balances precision and recall for this screening exercise.

Table 2. Validation model comparison

Model	Train PR-AUC	Validation PR-AUC	Gap	Validation ROC-AUC
Random Forest	0.459	0.337	0.122	0.895
Decision Tree	0.325	0.252	0.073	0.921
Logistic Regression	0.162	0.159	0.003	0.836
Dummy	0.039	0.042	-0.003	0.500

Random Forest had the highest validation PR-AUC at 0.337 and was therefore selected. The Decision Tree had a higher validation ROC-AUC, but PR-AUC was chosen in advance as the primary metric because the positive class is rare. Random Forest also showed a larger train-validation gap, so I checked generalization rather than judging the model only by training performance.

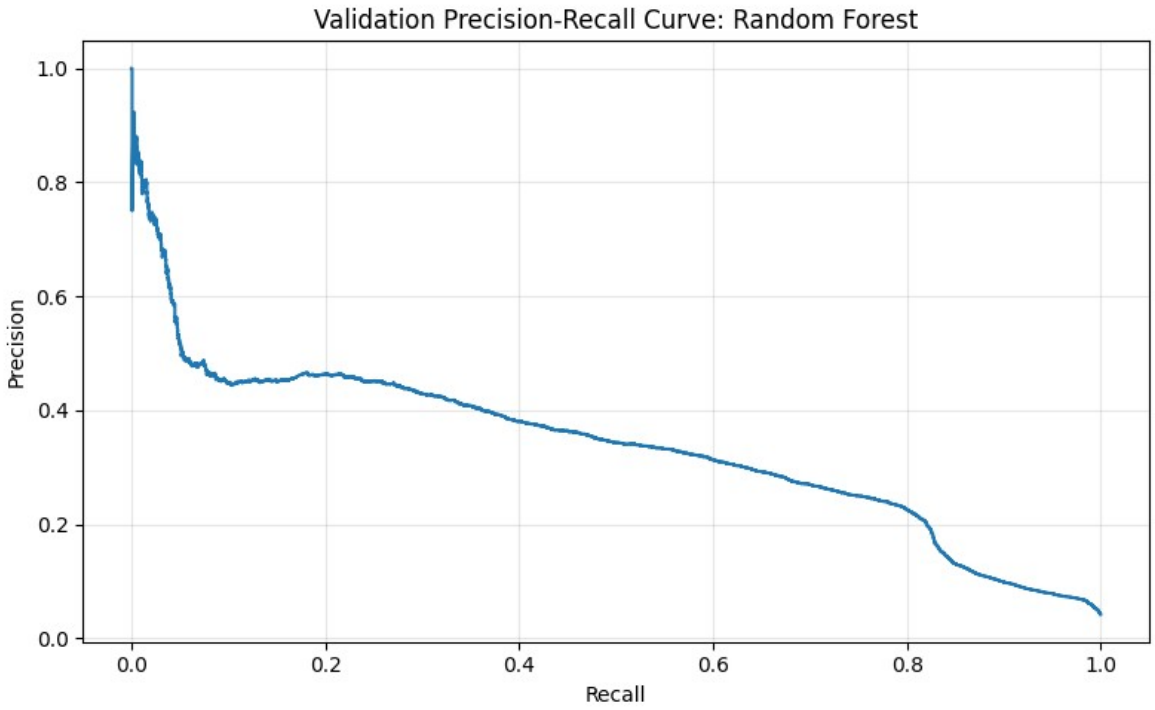


Figure 2. Validation precision-recall curve for the selected Random Forest.

5. Validation Threshold and Generalization

The selected validation threshold was 0.690. At this threshold, validation precision was 0.332, recall was 0.557, and F1 was 0.416. The validation confusion matrix contained 87,347 true negatives, 4,496 false positives, 1,776 false negatives, and 2,234 true positives. A false positive means that the screening model flags a loan as high risk even though it does not charge off within five years; a false negative means that the model fails to flag a loan that later charges off.

The learning-curve diagnostic showed evidence of some overfitting. The train-validation PR-AUC gap fell from about 0.218 with 25,000 training observations to 0.174 with 100,000. Validation PR-AUC ended higher overall, increasing from 0.285 to 0.309, although the improvement was not monotonic at every sample size. This suggests that additional data can help, but the selected Random Forest still fits the training data more strongly than the validation data.

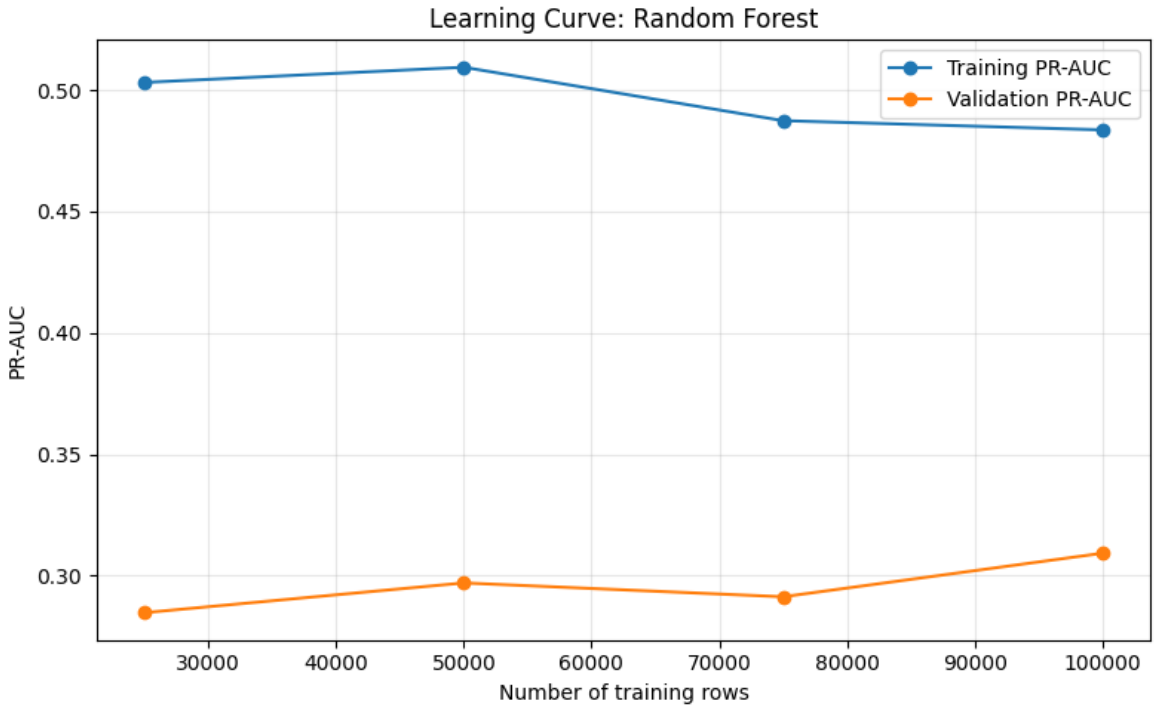


Figure 3. Learning-curve diagnostic for the selected Random Forest.

6. Final Test Results

After feature choices, model selection, and threshold selection were complete, I combined the training and validation samples, refit the selected Random Forest, and evaluated it on the 2020 through June 2021 test period. The final test set was not used to choose the model or threshold.

Table 3. Final locked-test performance

Metric	Value
PR-AUC	0.352
ROC-AUC	0.931
Precision	0.426
Recall	0.314
F1	0.362
Threshold	0.690
Test rows	54,249
Test charge-off rate	2.36%

The test set contained 1,280 actual five-year charge-offs. The model correctly identified 402 and missed 878. It also produced 542 false positives. In practical terms, recall of 0.314 means the model identified about 31.4% of actual charge-offs at the selected threshold, while precision of 0.426 means about 42.6% of flagged loans actually charged off within five years.

The test charge-off base rate was only 2.36%, so this remained a highly imbalanced problem. A PR-AUC of 0.352 is substantially above that positive-class base rate, which indicates useful ranking information even though the selected threshold captures only part of the charge-off population. The ROC-AUC of 0.931 is high, but the lower precision and recall values show why ROC-AUC alone would give an incomplete view of performance in this setting.

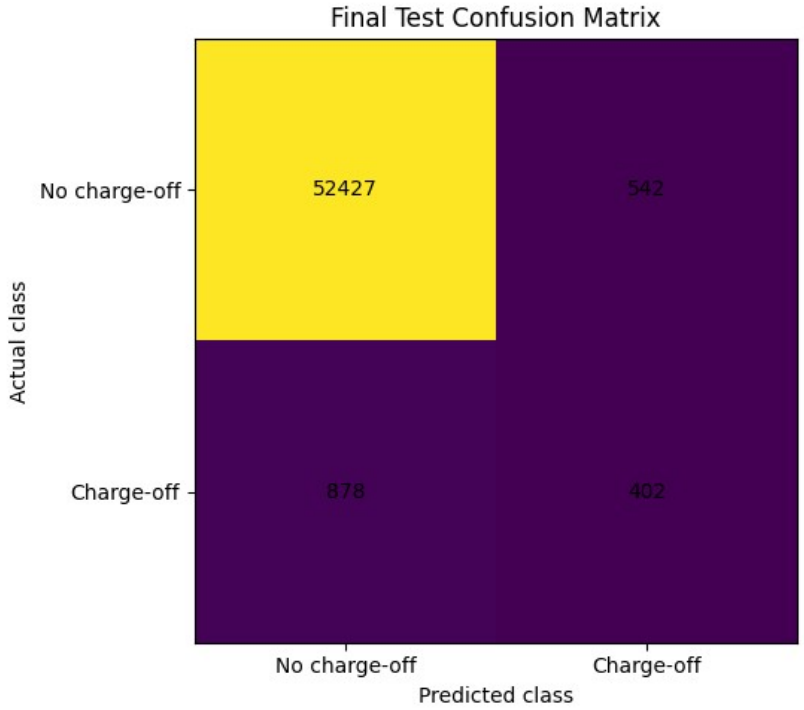


Figure 4. Final test confusion matrix at the validation-selected threshold.

7. Error Analysis and Model Interpretation

Validation error analysis was used to understand which loans the model classified incorrectly without using the final test set to make model changes. The most confident false negatives were not concentrated in a single state or industry. Several were relatively large or long-term loans, and many had a 75% SBA guarantee. Broad NAICS sectors 48, 42, 71, and 23 were among the sectors with the highest validation error rates when the comparison was limited to sectors with at least 100 validation loans. I treat these patterns as descriptive evidence about where the model struggles rather than as causal statements about industry risk.

For the final Random Forest, loan term was by far the largest feature importance at about 0.594. Initial interest rate (0.078), log loan amount (0.076), and SBA guarantee percentage (0.034) were the next largest numeric features. This broadly matches the EDA, where shorter terms, smaller loans, and higher interest rates were associated with higher observed charge-off rates. Random Forest importance measures usefulness to this fitted model; it does not measure a causal economic effect.

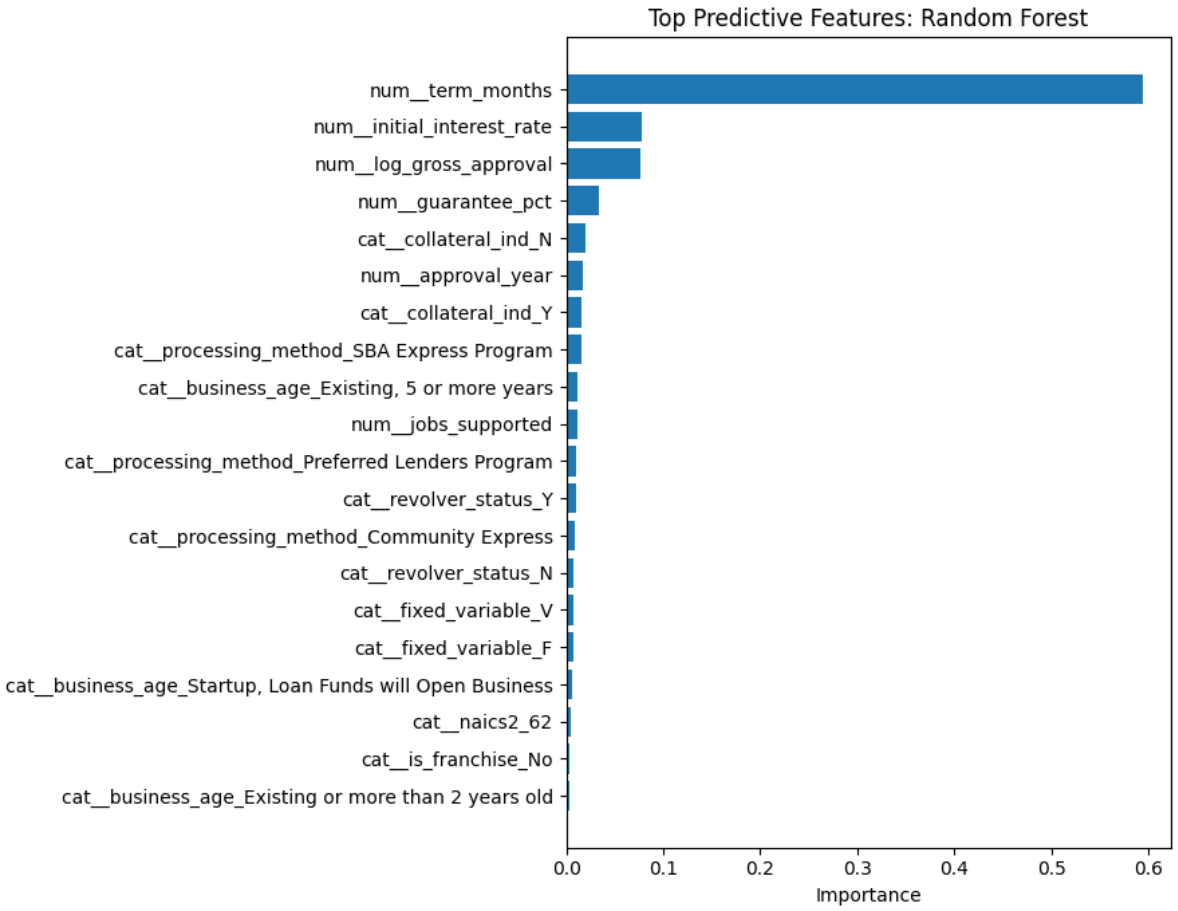


Figure 5. Top predictive features from the final Random Forest.

8. Stakeholders and Practical Interpretation

The results are most useful as a screening or portfolio-ranking exercise. A lender could use a model like this to prioritize files for additional review or to study which parts of a portfolio

deserve more monitoring. However, the final threshold should not be interpreted as a recommended underwriting cutoff. In a real bank, the cost of missing a risky loan would need to be compared with the cost of additional review, and the threshold would normally be chosen using explicit economic or policy objectives rather than F1 alone.

The analysis also matters for borrowers and SBA program stakeholders because false positives and false negatives have different consequences. A false positive could subject a performing loan to unnecessary extra scrutiny, while a false negative represents credit risk that the screening model failed to identify. This makes error analysis and transparency about limitations important even when overall ranking metrics are strong.

9. Limitations

- The public SBA files do not contain the full borrower financial information normally used in commercial underwriting, including cash flow, DSCR, leverage, liquidity, detailed credit history, management quality, or complete financial statements.
- Loans approved after June 30, 2021 are excluded because the June 30, 2026 snapshot does not provide a full five-year outcome window for those loans. The model is therefore not evaluated on the most recent SBA cohorts.
- The final test period covers 2020 through June 2021 and includes COVID-era originations. Differences between validation and test performance may partly reflect changes in the economic and policy environment.
- Charge-off is an observable lending outcome but is not identical to every definition of default, delinquency, restructuring, or financial distress.
- The sample contains approved SBA 7(a) loans. Rejected applicants are not observed, so the results describe risk among approved SBA loans rather than all small-business credit applicants.
- The analysis is predictive rather than causal. Relationships between variables such as interest rate or loan term and charge-off should not be interpreted as proof that changing those variables would change default behavior.
- The validation threshold is chosen by maximizing F1 for the course project. It is not an economically optimized bank cutoff.
- The data-quality check identified 208 training loans with non-positive terms. This is a very small share of the training set, but it remains a data-quality limitation.

10. Conclusion

This project finds that information available in the public SBA 7(a) files around loan approval contains meaningful predictive signal for five-year charge-off risk. Random Forest produced the strongest validation PR-AUC and achieved a test PR-AUC of 0.352 and ROC-AUC of 0.931 on the later 2020-June 2021 cohort. At the validation-selected threshold, the model achieved 42.6% precision and 31.4% recall on the test set.

The model is clearly better suited to ranking and screening than to making a stand-alone lending decision. It misses many actual charge-offs at the chosen threshold, and the public data omit important borrower-level financial information that would normally be available to a lender. The

main value of the project is therefore not a claim that public SBA variables are enough for underwriting, but evidence that they do provide useful risk information when the outcome is defined carefully, leakage is controlled, and performance is evaluated on later loan cohorts.

References

[1] U.S. Small Business Administration. 7(a) & 504 FOIA. SBA Open Data. https://data.sba.gov/dataset/7a-504-foia

[2] Géron, Aurélien. Hands-On Machine Learning with Scikit-Learn and PyTorch. O'Reilly Media. Companion repository: https://github.com/ageron/handson-mlp

[3] DATA 400 course materials, Fall 2026. End-to-End ML Project and Machine Learning Fundamentals lectures.

[4] Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel, O., et al. (2011). Scikit-learn: Machine Learning in Python. Journal of Machine Learning Research, 12, 2825-2830.